

E. Good Triples

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Given a non-negative integer number n ($n \geq 0$). Let's say a triple of non-negative integers (a, b, c) is good if $a + b + c = n$, and $\text{digsum}(a) + \text{digsum}(b) + \text{digsum}(c) = \text{digsum}(n)$, where $\text{digsum}(x)$ is the sum of digits of number x .

For example, if $n = 26$, then the pair $(4, 12, 10)$ is good, because $4 + 12 + 10 = 26$, and $(4) + (1 + 2) + (1 + 0) = (2 + 6)$.

Your task is to find the number of good triples for the given number n . The order of the numbers in a triple matters. For example, the triples $(4, 12, 10)$ and $(10, 12, 4)$ are two different triples.

Input

The first line of input contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases. Descriptions of test cases follow.

The first and only line of the test case contains one integer n ($0 \leq n \leq 10^7$).

Output

For each test case output one integer, the number of good triples for the given integer n . Order of integers in a triple matters.

Example

input	output
12	9
11	1
0	3
1	6
2	10
3	15
4	21
5	1350
3141	166375
999	29160
2718	1522435234375
9999999	3
10000000	

Note

In the first example, the good triples are $(0, 0, 11)$, $(0, 1, 10)$, $(0, 10, 1)$, $(0, 11, 0)$, $(1, 0, 10)$, $(1, 10, 0)$, $(10, 0, 1)$, $(10, 1, 0)$, $(11, 0, 0)$.

In the second example, there is only one good triple $(0, 0, 0)$.